

Design and Neutronics of a Subcritical-Assembly Simulator for Reactor-Physics Education

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ABSTRACT

We report the current development status of a full-scale physical simulator of a source-driven subcritical assembly for reactor-physics education where zero-power training facilities are unavailable. The baseline design is AGN-201-inspired and emulates a compact UO₂–polyethylene fuel system with two movable fuel rods and a movable AmBe source capsule. To support real-time operation on a single-board computer (SBC), the physics engine follows a quasi-static two-step approach: OpenMC pre-computations provide kinetics parameters and discrete mappings from the physical state to k_{eff} and source importance, with ongoing refinement to reduce statistical uncertainty for continuous-map fitting, and an online point-kinetics solver uses these mappings during operation. The neutronics results show $k_{\text{src}} < k_{\text{eff}}$ across all evaluated configurations, with k_{src} approaching k_{eff} as $k_{\text{eff}} \rightarrow 1$, highlighting an important teaching point for subcritical experiments. Implementation results include end-to-end SBC electronics/software operation (sensor acquisition, state-to-neutronics mapping, real-time kinetics, and web output), with detector responses modelled in software and reactor kinetics available in deterministic point-kinetics and stochastic Hawkes-process modes; the remaining hardware work is finalisation of the 1:1 physical-interface design, followed by assembly. The platform targets hands-on exercises including approach to critical, source/rod jerk, and rod oscillation, with planned extension to neutron-noise methods, enabling authentic reactor-physics laboratories without licensing or siting burdens.

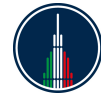
Keywords: subcritical assembly, source-driven systems, reactor-physics education, Monte Carlo, point kinetics

1. INTRODUCTION

The outlook for global nuclear capacity continues to strengthen [1]. In step with this trend, universities and professional bodies have launched or expanded nuclear-engineering BSc/MSc programmes [2, 3]. Yet the gradual disappearance of zero-power training facilities (ZPTFs)—a term we use here to encompass both low-power critical reactors and subcritical assemblies—has sharply limited opportunities to teach applied reactor physics through direct experimentation [4, 5]. Because ZPTFs are typically simple to operate, relatively accessible, and possess favourable inherent safety characteristics, they allow students to observe reactor-physics phenomena first-hand, translating abstract classroom lessons into durable understanding [6]. Recognising this educational value, several initiatives have sought to restore hands-on capacity, including the recommissioning of the MIT Graphite Exponential Pile and the commissioning of the VR-2 at the Czech Technical University; both subcritical ZPTFs [7, 8].

Despite such efforts, establishing new facilities remains slow, costly, and uncommon. Many countries have experienced prolonged hiatuses in licensing and constructing new reactors [9]. Even non-power facilities typically undergo multi-year regulatory reviews [10] and face recurring licensing fees during operation. Institutions may also hesitate to invest in new ZPTFs given the perception that reactor physics is a mature field and that high-fidelity modelling addresses much foundational work; these facilities are thus often positioned primarily as platforms for code and data validation rather than broad

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discovery [4, 5]. The VR-2 timeline illustrates these barriers: approximately nine years from project initiation to commissioning, including several years of regulatory review [8]. The result is a widening gap between growing educational demand and the shrinking availability of accessible, hands-on training environments.

To address this gap, we propose an interactive, full-scale physical simulator designed for reactor-physics education. The system couples a 1:1 physical interface, enabling realistic, hands-on interaction, with a quasi-static point-kinetics physics engine leveraging data pre-computed via Monte Carlo simulations. This approach captures essential phenomena and operational responses, making it well suited to classroom instruction. In the near term, the simulator can stand in for ZPTFs for the targeted exercises, offering comparable instructional value without licensing requirements, special containment, or high capital costs. By lowering logistical and financial barriers, it enables universities to integrate laboratory exercises into their curricula rapidly and at scale.

This paper covers the status of the development work on the proposed simulator. Section 2 presents our design choice and gives the underlying motivation, while Section 3 showcases selected neutronics calculations done in the support of the simulator's development. Section 4 then details its physical implementation and Section 5 briefly discusses the laboratory exercises enabled by the simulator. Finally, Section 6 presents conclusions and future outlook.

2. DESIGN RATIONALE & CHARACTERISTICS

In the selection process to establish the baseline design of the physical simulator, the objective was to choose a system for which a 1:1 scale physical representation can be built and with which students can directly interact, while a real-time computational engine simulates its physics behaviour. The overarching goal was to preserve the realism of operating an actual training reactor, while satisfying the following simulator requirements:

Small size, i.e. limited notional shielding required. This was favoured for practicality: if the device is envisioned to be transportable, for example to be exhibited as a public outreach tool, then the size needs to be limited. Furthermore, it is advantageous if the students can reach all parts of the device without any support such as stands or ladders, and if they also have the possibility to assemble the system by hand.

Manipulation by hand. This we consider to be of utmost importance, since in nuclear engineering education, students seldom have the possibility for hands-on operation. Such practice can teach students about safety principles while handling nuclear material, even when it is emulated with mock pieces. The hand-manipulation requirement makes it unrealistic to consider a system which can notionally become supercritical, or even critical; thus, we opted for a subcritical configuration.

Off-the-shelf hardware. The manipulation of the state of the system needs to be captured and processed using standard electronics components, without the need for a high-performance computer. To make the simulator self-contained, it was decided to deploy the computational engine emulating its physics response on a single-board computer (SBC) integrated within the physical interface and interfacing with input sensors. This choice of hardware, combined with the need for processing the physics response of the system *in situ* and in real time, complicates or potentially even precludes the possibility to use high-fidelity neutronics. Thus, we opted for a quasi-static, two-step approach. *First*, the neutronic state of the system (e.g. its multiplication factor) is pre-computed off-line with a Monte Carlo code for a sufficiently high number of physical configurations, allowing in principle for the construction of a continuous mapping. *Second*, this mapping is incorporated into the deployed physics engine, which executes point kinetics in real time, while continuously adapting the model parameters to the state of the physical system fetched by the input sensors.

The above requirements led us to survey UO₂-polyethylene dispersion systems as the basis for the simulated design. Several zero-power training facilities of this type have been built and operated, no-

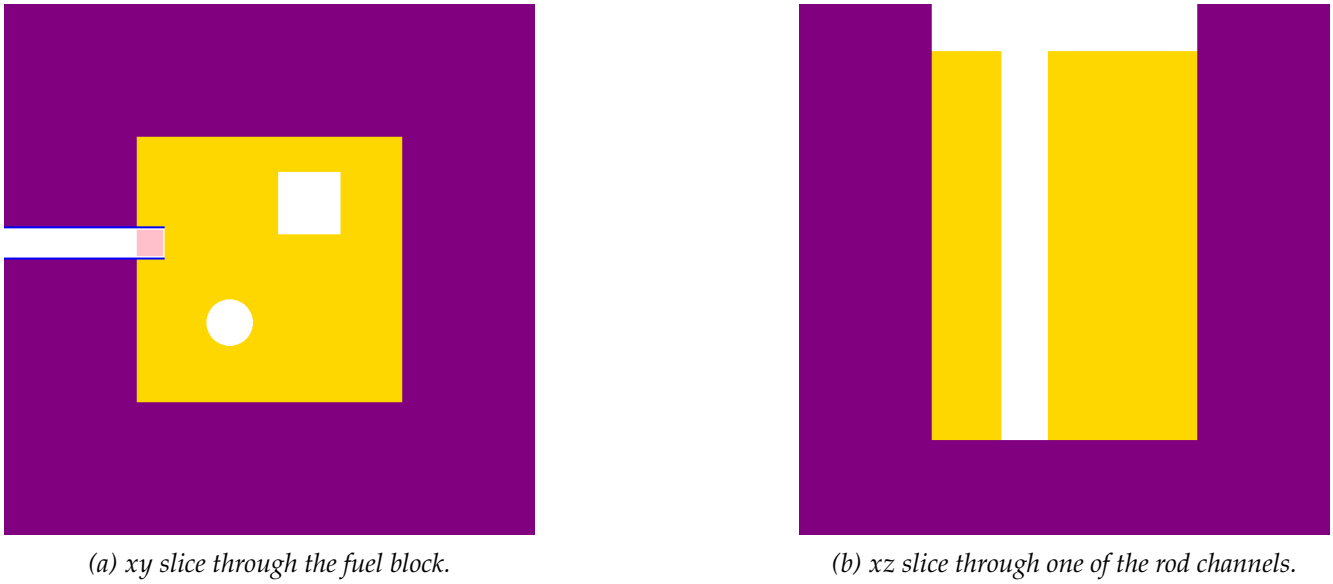


Figure 1. Orthogonal cuts through the modelled geometry coloured by material. Fuel rods withdrawn.

Table I. Kinetic parameters computed at full rod insertion. The uncertainties in λ and ℓ_p are negligible; the componentwise 1σ standard uncertainties in β are 0.5-1.5 pcm. Additionally, $\beta_{\text{eff}} = \sum_i \beta_i$.

Group i	1	2	3	4	5	6	β_{eff} (pcm)	ℓ_p (μs)
β_i (pcm)	27.3	139.3	133.5	292.4	123.8	51.6	767.9	42.8
λ_i (s^{-1})	1.33×10^{-2}	3.27×10^{-2}	1.21×10^{-1}	3.03×10^{-1}	8.50×10^{-1}	2.85×10^0		

tably the Siemens SUR-100 training reactors [11], the Hungarian experimental reactor ZR-4 [12], and the Aerojet General Nucleonics AGN-201 teaching reactors [13]. These facilities share attributes aligned with our goals, such as compact geometry and relative simplicity of construction and operation. Among them, the AGN-201 was particularly attractive because its design and performance are comprehensively documented in the open literature [14, 15, 16], and prior work has even proposed a training *subcritical* assembly re-using AGN-201 fuel plates [16].

Building on the AGN-201 documentation, we opted for a simple design composed of a parallelepiped fuel block surrounded by a graphite reflector. An external neutron source can be inserted via a dedicated channel, while the reactivity of the system is manipulated via the movement of horizontal fuel rods. The present geometry of the baseline design is illustrated in Figs. 1a and 1b, where yellow indicates the UO_2 -polyethylene fuel, purple the graphite reflector, pink the AmBe neutron source capsule, and blue the polyethylene source channel tubing.

3. NEUTRONICS MODELLING

This section showcases some of the neutronics calculations done during the development of the aforementioned two-step approach for reactor physics simulation. To obtain the data needed for point-kinetics simulations, the design was modelled using the OpenMC Monte Carlo code [17] with the official ENDF/B-VII.1 data libraries. The calculated kinetics parameters, i.e. the delayed-neutron precursor (DNP) decay constants λ (1/s) and fractions β (-) as well as the prompt neutron lifetime ℓ_p (s), are compiled in Table I. Six DNP groups were used. Since the possibility to calculate β and ℓ_p was added to OpenMC only recently [18], the development version v0.15.3-dev133 was used for this evaluation. The calculated β and λ agree well with the AGN-201 benchmark data obtained with the McCARD code

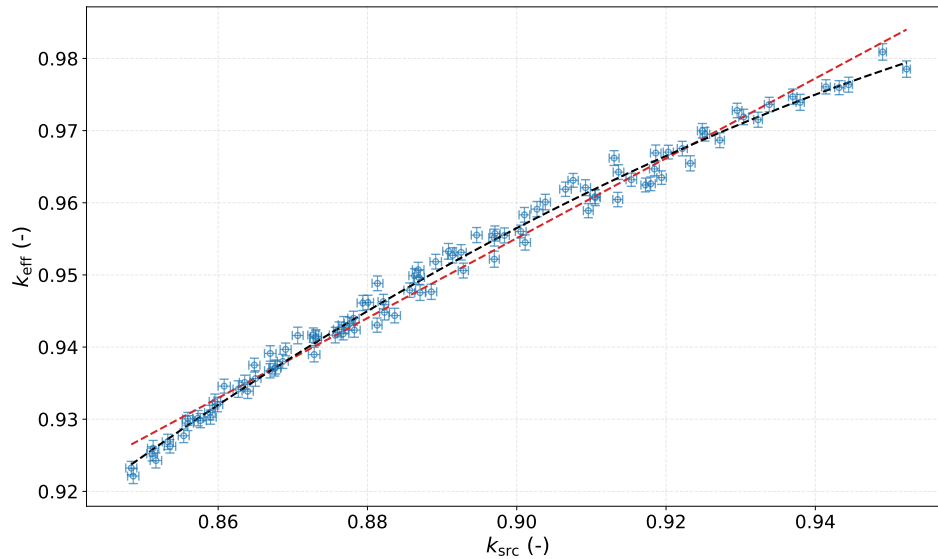


Figure 2. Multiplication coefficient k_{eff} vs source-driven multiplication k_{src} with 1σ uncertainty bars. Linear (dashed red) and quadratic (dashed black) lines of best fit are also shown.

[15], while the prompt neutron lifetime significantly disagrees. The reason for the discrepancy could be attributed to the additional graphite reflector as well as the lead and water shields present in the AGN-201 benchmark. Our value is however in good agreement with values of ℓ_p reported elsewhere for the AGN-201 [19, 20].

As mentioned above, the point-kinetics simulator also requires the pre-computation of the mapping from the physical system state x , here corresponding to the positions of the fuel rods and the neutron source capsule, to its neutronics state parameters, in particular the multiplication factor k_{eff} (-). Additionally, due to the spatial dependence of neutron importance and its variation during the approach to criticality, the external neutron source strength S_0 (n/s) must be augmented by the source importance ϕ^* (-) to obtain the effective source strength S_{eff} . Under subcritical conditions, ϕ^* can be recovered from fixed-source Monte Carlo calculations by requiring that the induced fission rate in the system equals the one derived from point kinetics for each system state x :

$$T(x) \cdot S_0 = \frac{k_{\text{eff}}(x) \cdot S_{\text{eff}}(x)}{1 - k_{\text{eff}}(x)}, \quad (1)$$

where T is the global nu-fission tally per source neutron calculated with OpenMC v0.15.2. Sometimes, source-driven subcritical multiplication k_{src} is introduced [21]. Then it can be derived that:

$$\phi^* = \frac{1 - k_{\text{eff}}}{k_{\text{eff}}} \frac{k_{\text{src}}}{1 - k_{\text{src}}}, \quad \text{where } k_{\text{src}} = \frac{T}{1 + T}. \quad (2)$$

The correspondence between k_{src} and k_{eff} across all evaluated rod positions is shown in Fig. 2. In every configuration, k_{src} is lower than the corresponding k_{eff} , reflecting the chosen source location and the associated neutron-importance weighting. A quadratic fit reproduces the trend slightly better than a linear one, but the linear one is more interpretable, with slope ≈ 0.55 . Thus k_{src} , which begins well below k_{eff} , approaches it as $k_{\text{eff}} \rightarrow 1$. This distinction matters for subcritical parameter estimation: experiments that rely on the external source, such as steady inverse multiplication or source-jerk tests, measure k_{src} , rather than k_{eff} [21]. We regard this as a further useful teaching point in the subcritical assembly simulator.

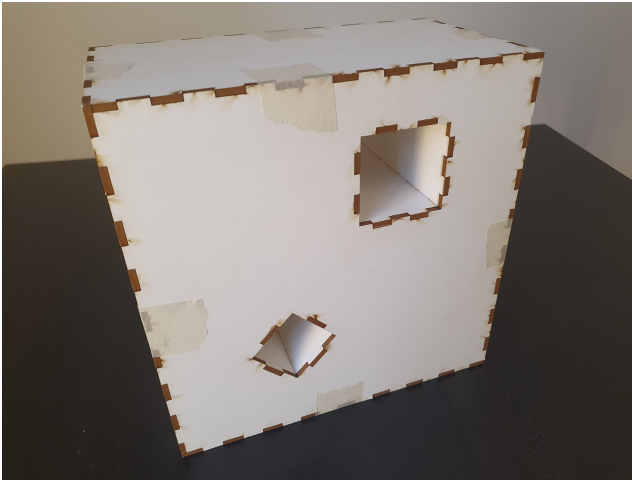


Figure 3. One of the early prototypes of the physical-interface components, in particular of a segment of the fuel block. Insertion channels for the two fuel rods are visible.

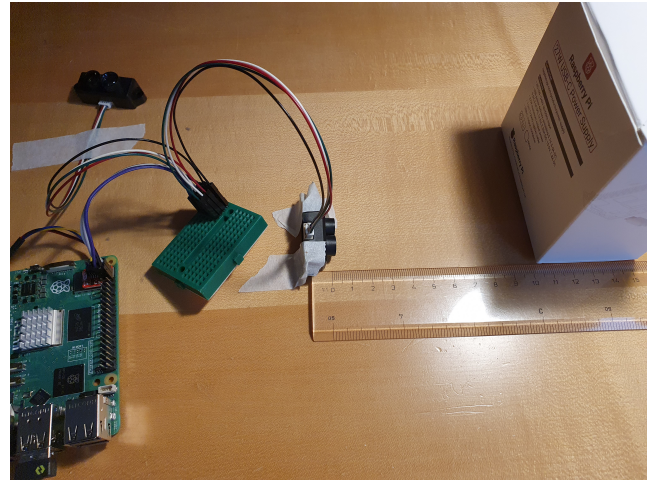


Figure 4. Test of a lidar sensor connected to a Raspberry Pi 5 SBC. Second lidar sensor with an unobstructed field of view is also visible in the photograph. Both sensors are connected to the SBC's I²C port.

Our present simulation focus is on computing the discrete mappings $x \rightarrow k_{\text{eff}}$ and $x \rightarrow \phi^*$ with lower statistical uncertainty, as this has been found to affect the efficacy of the fitting required for continuous map development. We also note that, while the DNP parameters discussed above are only weakly dependent on the rod positions, the prompt neutron lifetime was found to vary from $46.4 \mu\text{s}$ with both fuel rods withdrawn to $42.8 \mu\text{s}$ with both rods fully inserted. In future work, we will therefore also consider implementing an additional map for $x \rightarrow \ell_p$ to further improve the realism of the simulator's physics engine.

4. IMPLEMENTATION

The concrete implementation of the simulator comprises an instrumented physical interface and a real-time computational physics engine. The physical interface is designed for 1:1 correspondence with the modelled system and facilitates hands-on, real-world interaction. Figure 3 shows one of the early prototypes of its components, assembled from laser-cut cardboard plates. The final design of the physical interface is currently being finalised, after which the full-scale assembly will proceed. It will be instrumented via distance-measurement sensors, which will enable tracking the presence and degree of insertion of the mocked fuel rods and the external neutron source. Mock neutron detectors are currently not planned to be explicitly included in the physical interface; their location and detection characteristics are considered only implicitly in the computational physics engine. Note that for the fuel rods, lidar sensors will be employed for their good accuracy and small field of view, which enables their use for precise measurement of distances in enclosed channels.

The physics engine is implemented and executed in real time on a single-board computer (SBC) integrated into the simulator. The sensors relay their measurement data to the computer program, which maps measured distances to k_{eff} and S_{eff} using the pre-computed neutronics configuration data. These quantities drive the ongoing kinetics simulation, which advances in time and emits neutron counts per second for the modelled detectors. The results are displayed in an output web interface, which can be shown on a local screen connected to the SBC or accessed via a remote client. Figure 4 shows a test of a lidar distance sensor connected to a Raspberry Pi 5 SBC, while Fig. 5 presents representative output data from a rod-jerk "experiment". In this particular case, detector train characteristics are not considered; perfect detection is assumed and the displayed counts per second directly represent emulated neutron

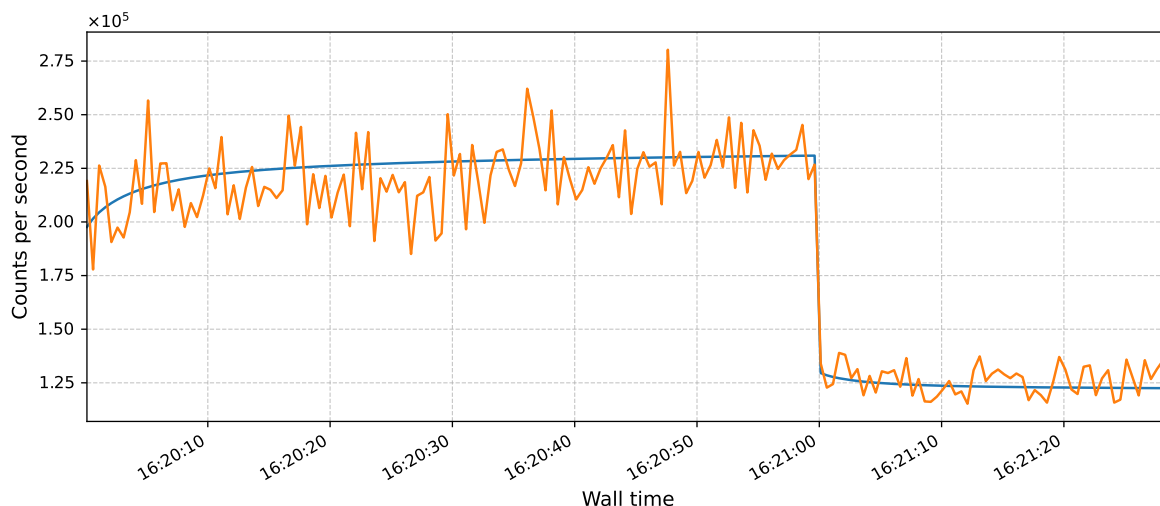


Figure 5. Output counts per second as a function of wall time during a rod-jerk "experiment". Blue: deterministic point-kinetics engine; orange: Hawkes process point-kinetics engine. Perfect detection is assumed.

birth events. Both the output of deterministic point kinetics and a stochastic Hawkes process model of the chain reaction [22] are shown.

As a summary, Fig. 6 shows a schematic diagram of the overall simulator implementation. At present, the electronics pipeline and associated software are complete, including sensor communication, real-time reactor-kinetics and detector-physics computation (with deterministic and stochastic options for reactor kinetics), and the output web interface. After the physical-interface design is finalised, the full simulator will be assembled and prepared for its intended use.

5. INTENDED LABORATORY EXERCISES

The value of interacting with a physical system cannot be overstated. Experience from radiation protection training shows that physically enacted procedures, even in fully safe mock environments, induce a mild but meaningful level of stress and attentional focus. This encourages greater care, procedural discipline, and adherence to safety boundaries than purely virtual simulations. Such training environments are well established in the nuclear industry, where mock-up facilities and procedural rehearsals are routinely used to prepare operators for real-world tasks. Our approach builds on this tradition by transferring these pedagogical benefits to reactor-physics education.

In spite of the simplicity of the presented simulator, it will be able to cover a significant fraction of typically-performed reactor-physics exercises:

1. *Approach to critical ($1/M$)* via gradual fuel rod insertion, followed by extrapolation to $k_{\text{eff}} = 1$.
2. *Source jerk and rod jerk* to estimate the subcritical reactivity.
3. *Rod oscillation* to observe the system's dynamic response.

Furthermore, we are exploring the use of the Hawkes process model to implement neutron noise exercises, emulating the Feynman- γ and Rossi- α experiments. The students will also compare their measured results with analytical calculations to better understand fundamental reactor-physics concepts such as buckling, leakage, and diffusion length.

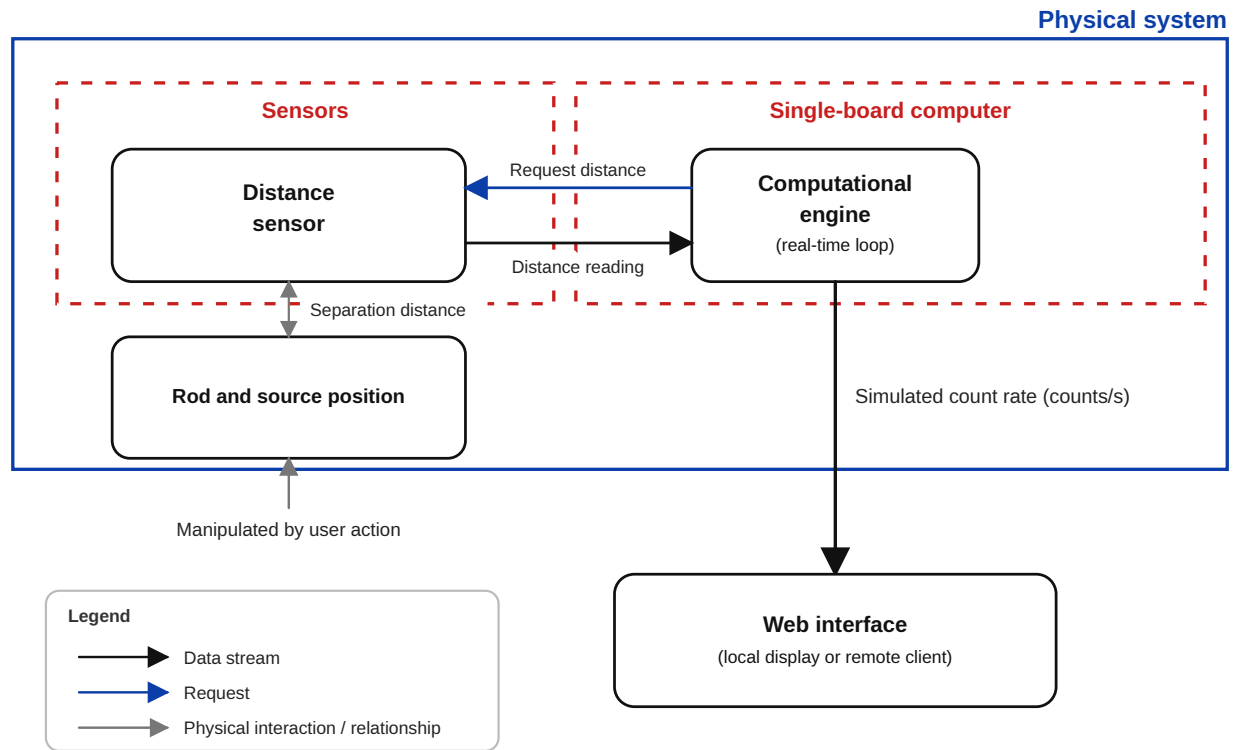


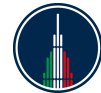
Figure 6. Schematic diagram of the overall simulator implementation.

6. CONCLUSIONS

In this paper, we reported the status of our development of a full-scale physical simulator for reactor-physics education. Its compact geometry, the envisioned hands-on two-fuel-rod/source-emulator interface, and the simple quasi-static governing equations support rapid development and classroom deployment. Next, we will finalise the remaining neutronics mappings and the physical-interface design, then assemble the full-scale mechanical interface. The resulting platform will enable authentic reactor-physics laboratories without licensing or siting burdens.

REFERENCES

- [1] International Atomic Energy Agency. "Energy, Electricity and Nuclear Power Estimates for the Period up to 2050." Technical Report IAEA-RDS-1/45, International Atomic Energy Agency, Vienna (2025).
- [2] Tennessee Tech University. "Tennessee Tech launches new degree program in nuclear engineering." (2024). URL www.tntech.edu/news/releases/24-25/tennessee-tech-launches-new-degree-program-in-nuclear-engineering.php.
- [3] M. Baldoni. "Polish Universities Launch New Courses To Support The Country's Nuclear Programme." (2023). URL www.euronuclear.org/news/polish-universities-launch-new-courses-to-support-the-countrys-nuclear-programme.
- [4] M. Margulis. "Zero power reactors in support of current and future nuclear power systems." *Nuclear Engineering and Design*, **volume 425**, p. 113330 (2024).
- [5] OECD Nuclear Energy Agency. "The demise of zero power reactors: From concern to action." (2023). URL www.oecd-neo.org/jcms/pl.79783/the-demise-of-zero-power-reactors-from-concern-to-action.



- [6] G. Wakabayashi, T. Yamada, T. Endo, and C. H. Pyeon. *Introduction to Nuclear Reactor Experiments*. Springer Singapore (2023).
- [7] M. D. Gale. *Developing modern graphite exponential pile experiments to augment reactor physics education*. Bachelor's thesis, Massachusetts Institute of Technology, Cambridge, MA (2018).
- [8] Czech Technical University in Prague. "The tenth nuclear reactor in the Czech Republic started operation at the Faculty of Nuclear Sciences and Physical Engineering of the Czech Technical University in Prague." (2023). URL aktualne.cvut.cz/en/press-reports/20230606-the-tenth-nuclear-reactor-in-the-czech-republic-started-operation-at-the.
- [9] OECD Nuclear Energy Agency. "Unlocking Reductions in the Construction Costs of Nuclear: A Practical Guide for Stakeholders." Technical Report NEA No. 7530, OECD/NEA (2020).
- [10] Nuclear Innovation Alliance. "Promoting Efficient NRC Advanced Reactor Licensing Reviews to Enable Rapid Decarbonization." Technical report, Nuclear Innovation Alliance (2021).
- [11] S. Radzuweit, M. Mielisch, K. D. Closs, and G. Zuber. "Options for the Management of Fuel from German Training Reactors." In *ENS RRFM '99: Transactions (Oral presentations and posters)*, pp. 134–138. 3rd International Topical Meeting on Research Reactor Fuel Management (RRFM '99), European Nuclear Society, Bruges, Belgium (1999).
- [12] F. Szabó. "The zero power reactor ZR-4." KFKI Report 72-48, Hungarian Academy of Sciences, Central Research Institute for Physics (MTA KFKI), Budapest (1972).
- [13] R. Davis, C. M. Perfetti, R. D. Busch, L. L. Wetzel, and C. A. Willis. "Research and educational applications of the Aerojet General Nucleonics 201-M at the University of New Mexico." *Annals of Nuclear Energy*, **volume 204**, p. 110564 (2024).
- [14] University of New Mexico, Department of Chemical and Nuclear Engineering. "AGN-201M Reactor: Reactor Operation and Training Manual." ADAMS record ML20195E222, Albuquerque, NM (1984).
- [15] J. W. Park and H. J. Park. "AGN-201K educational and research reactor benchmark analyses by McCARD Monte Carlo calculations." *Nuclear Engineering and Technology*, **volume 57**(3), p. 103234 (2025).
- [16] D. G. Bowen, M. Dupont, A. Lang, A. M. Holcomb, and S. Hart. "Proposed Subcritical Assembly for Nuclear Criticality Safety Training at the Oak Ridge National Laboratory." Technical report, Oak Ridge National Laboratory (2022).
- [17] P. K. Romano, N. E. Horelik, B. R. Herman, A. G. Nelson, B. Forget, and K. Smith. "OpenMC: A State-of-the-Art Monte Carlo Code for Research and Development." *Annals of Nuclear Energy*, **volume 82**, pp. 90–97 (2015).
- [18] J. Dorville, L. Labrie-Cleary, and P. K. Romano. "Implementation of the Iterated Fission Probability Method in OpenMC to Compute Adjoint-Weighted Kinetics Parameters." In *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (M&C 2025)*, pp. 984–994. American Nuclear Society, Denver, CO, USA (2025).
- [19] K.-O. Kim, H. J. Yoo, and B. W. Park. "Analysis of the Effective Neutron Generation Time Using the MCNP6 Code." In *Transactions of the Korean Nuclear Society (KNS) Virtual Autumn Meeting*. Online (2021).
- [20] J. S. Palmer, R. H. Stewart, and A. J. B. Shields. "Physics-Informed Recurrent Neural Networks to Predict Reactor Operations of the AGN-201 Nuclear Reactor." Technical Report INL/CON-25-82713-Revision-0, Idaho National Laboratory (2025).
- [21] International Atomic Energy Agency. "Considerations of Safety and Utilization of Subcritical Assemblies." Technical Report IAEA-TECDOC-1976, International Atomic Energy Agency, Vienna (2021).
- [22] L. Bureš. "Nuclear Chain Reaction Modelled via a Linear Hawkes Process." Submitted to the PHYSOR 2026 conference.